5 Interesting Reference Designs For BLDC MOTOR DRIVES

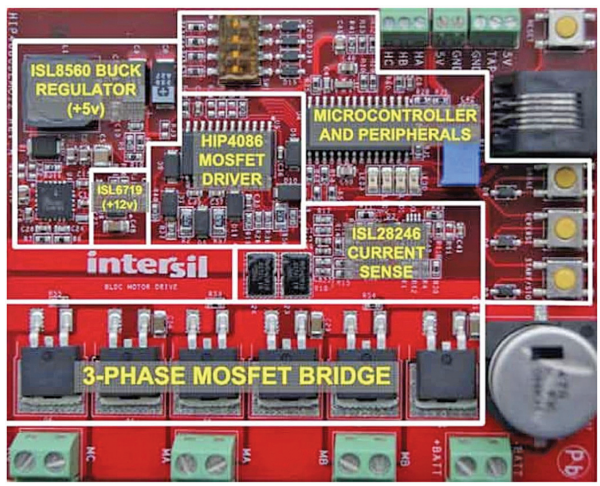
Explore reference designs for brushless direct current (BLDC) motor drives covering three-phase control, sensorless operation, dual-motor systems, and high-power motor control.

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BLDC motors appear everywhere, from fans and pumps to electric vehicles, drones, power tools, and industrial equipment. A good drive must control the three motor phases, detect rotor position, sense current, and handle protection, power conversion, and communication. Depending on the application, it may also need sensorless operation, field-oriented control (FOC), regenerative braking, or support for more than one motor. These five reference designs show how different original equipment manufacturers (OEMs) approach each need.

Three-Phase BLDC Motor Driver

A Hall-sensor board for trying out 15-60V motors without a closed-loop controller



Need to test a BLDC motor-drive design before building closed-loop control? Renesas's board combines a three-phase metal-oxide-semiconductor field-effect transistor (MOSFET) driver, a microcontroller, and Hall-effect sensors that detect rotor position. It supports 15V to 60V DC motors and a combined bridge current of up to 20A with sufficient airflow. Using pulse-width modulation (PWM) at 20kHz, it handles starting, stopping, reversing, braking, and open-loop speed control. Torque control, speed regulation, and regenerative braking are excluded, as these depend on the motor and load.

Takeaway. A starting point for BLDC drives in e-bikes, battery-powered tools, power steering, wheelchairs, appliances, e-mobility, and vehicle-control systems.

Link. *https://efy.co.in/fw3jel*

OEM Brand. Renesas

Sensorless BLDC Design

Rotor-position sensing without Hall sensors, on a 24V, 90W motor

Even without position sensors, the drive must still know where the rotor is. This NXP design estimates it from the motor's back-electromotive force (back-EMF) and uses it to switch the phases through six-step commutation. It adds closed-loop speed control, current limiting, and overvoltage, undervoltage, and overcurrent protection. Demonstrated with a 24V, 90W, 9350rpm three-phase motor, the board uses a 32-bit Arm Cortex-M0+ microcontroller and a field-effect transistor (FET) pre-driver that controls three high-side and three low-side switches. The pre-driver supports 5V and 3V logic signals. A system basis chip handles power management and sensor interfaces and adds a Controller Area Network (CAN) transceiver. For engineers, it shows how

